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Practical No. 10

Aim: To perform Hierarchical Clustering using agglomerative methods and interpret dendrograms for cluster formation.

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Aim: To perform **Hierarchical Clustering** using agglomerative methods and interpret dendrograms for cluster formation.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- It is effective on small datasets but behaves poorly on big datasets and with the spheroidal shape of the datasets..
- Understanding the concept of hierarchical clustering, implementing the algorithm, interpreting dendrograms, evaluating clustering results, handling different linkage methods, and applying hierarchical clustering to real-world datasets.
- As there are calculations involved in hierarchical clustering, its space and time complexity is very high
- Dendrograms are an integral part of hierarchical clustering, where the silhouette score is used to calculate the error in clustering algorithms.

THEORY:

What is Hierarchical Clustering?

Hierarchical clustering is an unsupervised learning algorithm used to group similar data points into clusters. It builds a multilevel hierarchy of clusters by either merging smaller clusters into larger ones (agglomerative) or dividing a large cluster into smaller ones (divisive). This results in a tree-like structure known as a dendrogram.
A dendrogram is a visual representation that illustrates the arrangement of clusters and their relationships with each other. The height of the branches in a dendrogram represents the distance or dissimilarity at which clusters merge. Lower heights indicate clusters joined at smaller distances, thus representing higher similarity.

Why hierarchical clustering matters in machine learning

Hierarchical clustering is particularly useful when the number of clusters is not known beforehand. It allows data analysts and data scientists to visually explore the data's structure through dendrograms.

This technique is also capable of uncovering nested clusters and is widely used in fields like genomics, customer segmentation, and document organization.
For example, let's say we have a dataset of customer purchase records. Using hierarchical clustering, we can group customers with similar purchasing patterns into clusters and identify potential market segments for targeted marketing strategies.

Hierarchical vs K-Means clustering

When it comes to clustering tasks, most are undecided on two main methods: Hierarchical clustering or K-means clustering.
But what makes them different? Let's look at their differences below for a clearer picture.
K-Means:

- Requires the number of clusters to be defined before running.
- Efficient and scalable for large datasets.
- Assumes clusters are spherical and equally sized.

Hierarchical Clustering:

- Does not require a predefined number of clusters.
- More interpretable due to dendrograms.
- Slower on large datasets but better for exploring structure.

When comparing the two, some areas make hierarchical clustering a stand-out option.
Here are some areas where hierarchical clustering is good:

- Flexibility in cluster selection using dendrograms.
- Good for small datasets or when a visual hierarchy is needed.
- Handles complex cluster shapes.

However, it can have some downsides, such as:

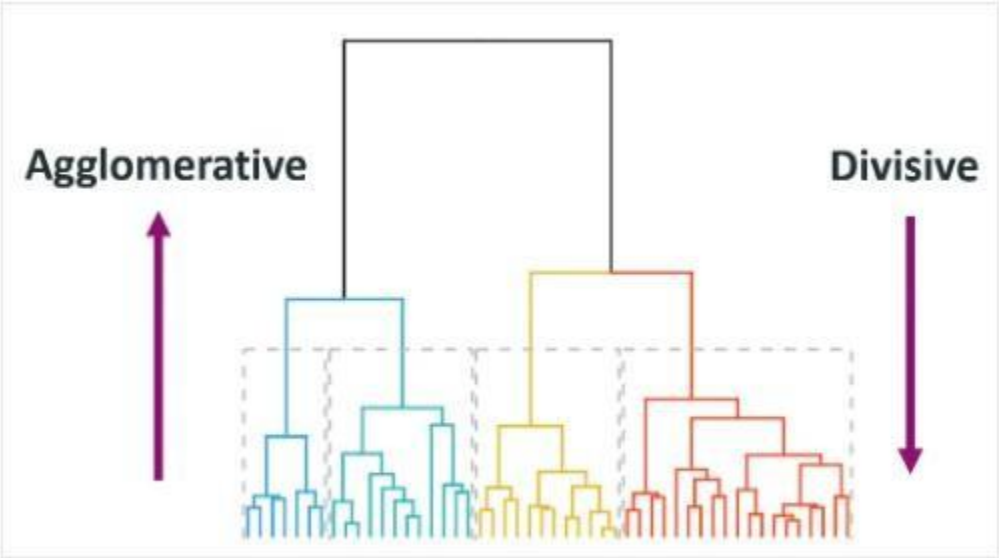
- Computationally expensive
- Sensitive to noisy data and outliers.

You might ask: When should I use hierarchical clustering?
Hierarchical clustering is an excellent choice in the following applications:

- Exploratory data analysis
- Gene expression pattern analysis
- Market segmentation with moderate-sized datasets

Types of Hierarchical Clustering

Within the technique of hierarchical clustering, you'll expect several types, with each providing different insights and results. The two main types of hierarchical clustering are agglomerative (bottom-up) and divisive (top-down).



Source: Himanshu Sharma on Medium

1. Agglomerative (Bottom-Up)

Agglomerative clustering starts with each data point as an individual cluster and iteratively merges the closest pair of clusters until only one cluster remains or until a stopping condition is met (like a desired number of clusters).
This method is also called bottom-up because it starts from the bottom (individual data points) and builds up to the top (final cluster).

Step-by-step guide:

1. Treat each data point as a singleton cluster.
2. Compute the pairwise distances between all clusters.
3. Merge the two clusters with the smallest distance.
4. Update the distance matrix to reflect the new set of clusters.
5. Repeat steps 2–4 until all points belong to a single cluster.
   This method is widely used due to its simplicity and ease of implementation.

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2. Divisive (Top-Down)

Divisive clustering starts with all data points in a single cluster and recursively splits them into smaller clusters. The process continues until each data point is in its own individual cluster.
It is also known as a top-down approach, as it starts at the top (single cluster) and breaks it down into smaller clusters.

Step-by-step guide:

1. Start with one large cluster containing all data points.

2. Split the cluster into two that are as different as possible.
3. Reapply the splitting process recursively to each resulting cluster.

Divisive methods are typically more computationally expensive due to their recursive nature, and accuracy depends greatly on the splitting strategy. Agglomerative methods are more common due to ease of implementation and widespread software support.

Advantages:

1. ***Performance:*** It is effective in data observation from the data shape and returns accurate results. Unlike KMeans clustering, here, better performance is not limited to the spheroidal shape of the data; data having any values is acceptable for hierarchical clustering.
2. ***Easy:*** It is easy to use and provides better user guidance with good community support. So much content and good documentation are available for a better user experience.
3. ***More Approaches:*** Two approaches are there using which datasets can be trained and tested, agglomerative and divisive. So if the dataset provided is complex and very hard to train on, we can use another approach.
4. ***Performance on Small Datasets:*** The hierarchical clustering algorithms are effective on small datasets and return accurate and reliable results with lower training and testing time.

Disadvantages:

1. ***Time Complexity:*** As many iterations and calculations are associated, the time complexity of hierarchical clustering is high. In some cases, it is one of the main reasons for preferring KMeans clustering.
2. ***Space Complexity:*** As many calculations of errors with losses are associated with every epoch, the space complexity of the algorithm is very high. Due to this, while implementing the hierarchical clustering, the space of the model is considered. In such cases, we prefer KMeans clustering.
3. ***Poor performance on Large Datasets:*** When training a hierarchical clustering algorithm for large datasets, the training process takes so much time with space which results in poor performance of the algorithms.

PROGRAM CODE:

CONCLUSION:

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DISCUSSION AND VIVA VOCE:

1. How would you use hierarchical clustering to segment customers, and why would you choose it over K-Means?
2. In a real project, how would you interpret a dendrogram to decide the optimal number of clusters?
3. Why is hierarchical clustering preferred for small datasets but not for large-scale industry applications?
4. How would you choose between single, complete, average, and Ward linkage methods for a business problem?
5. How can hierarchical clustering help in grouping patients based on medical records for personalized treatment?
6. How would you apply hierarchical clustering to group similar documents or news articles?
7. How do outliers affect hierarchical clustering, and how would you handle them in a real-world dataset?
8. In what situations would hierarchical clustering give better results than K-Means in industry?
9. How would you modify or approximate hierarchical clustering to handle large datasets?
10. After creating clusters using hierarchical clustering, how would you present and explain the results to non-technical stakeholders?

REFERENCE

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